

Module 3: Z-transforms

Introduction: Z-transform can be considered as a discrete-time equivalent of the Laplace transform. The Z-transform (ZT) is a mathematical tool which is used to convert the difference equations in time domain into the algebraic equations in z-domain.

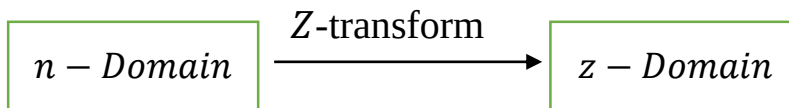
The Z-transform is a very useful tool in the analysis of a linear shift invariant (LSI) system. An LSI discrete time system is represented by difference equations. To solve these difference equations which are in time domain, they are converted first into algebraic equations in z-domain using the Z-transform, then the algebraic equations are manipulated in z-domain and the result obtained is converted back into time domain using the inverse Z-transform.

Definition: Let u_n be a function (sequence) defined for the discrete values $n = 0, 1, 2, \dots$, then Z-transform of u_n is defined as

$$Z[u_n] = \sum_{n=0}^{\infty} u_n z^{-n} = U(z) \quad [\text{provided the power series } z^{-n} \text{ converges}]$$

where n is a non negative integer and z is, in general, a complex number.

It follows that,



Remarks:

1. The bilateral (two sided) Z-transform of u_n is defined as

$$Z[u_n] = \sum_{n=-\infty}^{\infty} u_n z^{-n} = U(z).$$

2. The Z-transform does not converge for all sequences u_n or for all values of n .
3. Many sequences u_n will have Z-transform that can expressed in a compact form.
4. Note that

$$Z[u_n] = \sum_{n=0}^{\infty} u_n z^{-n} = u_0 + u_1 z^{-1} + u_2 z^{-2} + \dots = u_0 + \frac{u_1}{z^1} + \frac{u_2}{z^2} + \dots = U(z).$$

Linearity Property:

If a, b & c are constants and u_n, v_n & w_n are discrete functions, then

$$Z[au_n + bv_n - cw_n] = aZ[u_n] + bZ[v_n] - cZ[w_n]$$

NOTE:

- 1) $(1 - x)^{-1} = 1 + x + x^2 + x^3 + x^4 + \dots$ if $|x| < 1$
- 2) $(1 + x)^{-1} = 1 - x + x^2 - x^3 + x^4 - \dots$ if $|x| < 1$

3.1: Some Standard Z-transforms

1) $Z[a^n] = \frac{z}{z-a}$ & $Z[(-a)^n] = \frac{z}{z+a}$ where "a" is any constant.

Examples: $Z[2^n] = \frac{z}{z-2}$, $Z[-4^n] = -\frac{z}{z-4}$

$$Z\left[\frac{1}{3^n}\right] = Z\left[\left(\frac{1}{3}\right)^n\right] = \frac{z}{z-\frac{1}{3}}, \quad Z[(-2)^n] = \frac{z}{z+2}, \quad Z[(-4)^n] = \frac{z}{z+4}$$

2) $Z[1] = \frac{z}{z-1}$ & $Z[k] = \frac{kz}{z-1}$ where "k" is any constant

Examples: $Z[2] = \frac{2z}{z-1}$, $Z[3] = \frac{3z}{z-1}$, $Z[-4] = \frac{-4z}{z-1}$, $Z[0] = 0$

3) $Z[n^p] = -z \frac{d}{dz} [Z(n^{p-1})]$ where p is a positive integer.

i) $Z[n] = \frac{z}{(z-1)^2}$, ii) $Z[n^2] = \frac{z^2+z}{(z-1)^3}$, iii) $Z[n^3] = \frac{z^3+4z^2+z}{(z-1)^4}$

Examples: $Z[2n] = 2Z[n] = 2 \frac{z}{(z-1)^2} = \frac{2z}{(z-1)^2}$, $Z[-3n] = \frac{-3z}{(z-1)^2}$

$$Z[2n^2] = \frac{2(z^2+z)}{(z-1)^3}, \quad Z[-3n^2] = \frac{-3(z^2+z)}{(z-1)^3}$$

$$Z[2n^3] = \frac{2(z^3+4z^2+z)}{(z-1)^4}, \quad Z[-3n^3] = \frac{-3(z^3+4z^2+z)}{(z-1)^4}$$

4) i) $Z[\cos n\theta] = \frac{z^2 - z \cos \theta}{z^2 - 2z \cos \theta + 1}$, ii) $Z[\sin n\theta] = \frac{z \sin \theta}{z^2 - 2z \cos \theta + 1}$

Examples: $Z[\cos 2n] = \frac{z^2 - z \cos 2}{z^2 - 2z \cos 2 + 1}$, $Z[\sin 3n] = \frac{z \sin 3}{z^2 - 2z \cos 3 + 1}$

5) i) $Z[\cosh n\theta] = \frac{z^2 - z \cosh \theta}{z^2 - 2z \cosh \theta + 1}$, ii) $Z[\sinh n\theta] = \frac{z \sinh \theta}{z^2 - 2z \cosh \theta + 1}$

Examples: $Z[\cosh 2n] = \frac{z^2 - z \cosh 2}{z^2 - 2z \cosh 2 + 1}$, $Z[\sinh 3n] = \frac{z \sinh 3}{z^2 - 2z \cosh 3 + 1}$

Damping Rules:

$$\text{i) } Z[a^n u_n] = [Z(u_n)]_{z \rightarrow \frac{z}{a}}$$

$$\text{ii) } Z[a^{-n} u_n] = [Z(u_n)]_{z \rightarrow az}$$

$$1) Z[a^n n] = \frac{az}{(z-a)^2}$$

$$\text{Examples: } Z[2^n n] = \frac{2z}{(z-2)^2}, \quad Z[(-4)^n n] = \frac{-4z}{(z+4)^2}$$

$$2) Z[a^n n^2] = \frac{az^2 + a^2 z}{(z-a)^3}$$

$$\text{Examples: } Z[2^n n^2] = \frac{2z^2 + 4z}{(z-2)^3}, \quad Z[(-4)^n n^2] = \frac{-4z^2 + 16z}{(z+4)^3}$$

$$3) Z[a^n \cos n\theta] = \frac{z^2 - az \cos \theta}{z^2 - 2az \cos \theta + a^2}$$

$$\text{Examples: } Z[2^n \cos 3n] = \frac{z^2 - 2z \cos 3}{z^2 - 4z \cos 3 + 4}, \quad Z[(-3)^n \cos 2n] = \frac{z^2 + 3z \cos 2}{z^2 + 6z \cos 2 + 9}$$

$$4) Z[a^n \sin n\theta] = \frac{az \sin \theta}{z^2 - 2az \cos \theta + a^2}$$

$$\text{Examples: } Z[2^n \sin 3n] = \frac{2z \sin 3}{z^2 - 4z \cos 3 + 4}, \quad Z[(-3)^n \sin 2n] = \frac{-3z \sin 2}{z^2 + 6z \cos 2 + 9}$$

PROBLEMS

$$1) \text{ Find the } Z\text{-transform of } 3n - 4 \sin\left(\frac{n\pi}{4}\right) + 5a.$$

$$\text{Answer: } Z\left[3n - 4 \sin\left(\frac{n\pi}{4}\right) + 5a\right] = \frac{3z}{(z-1)^2} - \frac{2\sqrt{2}z}{z^2 - \sqrt{2}z + 1} + \frac{5az}{(z-1)}$$

$$2) \text{ Find the } Z\text{-transform of } (n+1)^2.$$

$$\text{Answer: } Z[(n+1)^2] = \frac{z^2(z+1)}{(z-1)^3}$$

$$3) \text{ Find the } Z\text{-transform of } 4(3)^n + 2(-1)^n.$$

$$\text{Answer: } Z[4(3)^n + 2(-1)^n] = 4\left[\frac{z}{z-3}\right] + 2\left[\frac{z}{z+1}\right] = \frac{4z}{z-3} + \frac{2z}{z+1}$$

$$4) \text{ Find the } Z\text{-transform of } \sin\left(\frac{n\pi}{2}\right).$$

$$\text{Answer: } Z\left[\sin\left(\frac{n\pi}{2}\right)\right] = \frac{z}{z^2+1}$$

5) Find the Z -transform of $\cos\left(\frac{n\pi}{2}\right)$.

Answer: $Z\left[\cos\left(\frac{n\pi}{2}\right)\right] = \frac{z^2}{z^2+1}$

6) Find the Z -transform of $\sin(3n + 5)$.

Answer: $Z[\sin(3n + 5)] = \frac{z \sin 3 \cos 5 + \sin 5 (z^2 - z \cos 3)}{z^2 - 2z \cos 3 + 1}$

7) Find the Z -transform of $\cos\left(\frac{n\pi}{2} + \frac{\pi}{4}\right)$.

Answer: $Z\left[\cos\left(\frac{n\pi}{2} + \frac{\pi}{4}\right)\right] = \frac{1}{\sqrt{2}} \left[\frac{z^2 - z}{z^2 + 1} \right] = \frac{z}{\sqrt{2}} \left[\frac{z-1}{z^2+1} \right]$

8) Find the Z -transform of ne^{an} .

Answer: $= \frac{e^a z}{(z - e^a)^2}$

9) Find the Z -transform of $e^n \sin 2n$.

Answer: $= \frac{ez \sin 2}{z^2 - 2ez \cos 2 + e^2}$

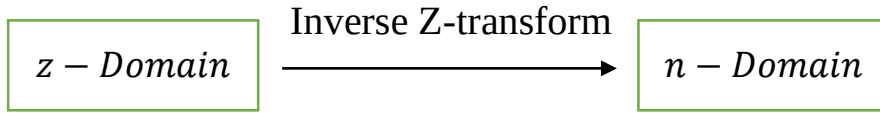
10) Find the Z -transform of $n \sin n\theta$.

Answer: $Z[n \sin n\theta] = \frac{z \sin \theta (z^2 - 1)}{(z^2 - 2z \cos \theta + 1)^2}$

3.2: Inverse Z-transforms

Definition: If $Z[u_n] = U(z)$, then $Z^{-1}[U(z)] = u_n$ is called inverse Z-transform.

Clearly,



Linearity Property:

If a, b & c are constants and $U(z)$, $V(z)$ & $W(z)$ are functions of z , then

$$Z^{-1}[aU(z) + bV(z) - cW(z)] = aZ^{-1}[U(z)] + bZ^{-1}[V(z)] - cZ^{-1}[W(z)].$$

List of Standard Inverse Z-transforms:

1) $Z^{-1}\left[\frac{z}{z-1}\right] = 1$

Examples: $Z^{-1}\left[\frac{2z}{z-1}\right] = 2$, $Z^{-1}\left[\frac{-3z}{z-1}\right] = -3$, $Z^{-1}\left[\frac{\pi z}{z-1}\right] = \pi$

2) $Z^{-1}\left[\frac{z}{z-a}\right] = a^n$ & $Z^{-1}\left[\frac{z}{z+a}\right] = (-a)^n$

Examples: $Z^{-1}\left[\frac{z}{z-2}\right] = 2^n$, $Z^{-1}\left[\frac{z}{z+4}\right] = (-4)^n$

$$Z^{-1}\left[\frac{z}{z-\frac{1}{2}}\right] = \left(\frac{1}{2}\right)^n; \quad Z^{-1}\left[\frac{2z}{z+\frac{1}{3}}\right] = 2\left(-\frac{1}{3}\right)^n$$

3) $Z^{-1}\left[\frac{z}{(z-1)^2}\right] = n$, $Z^{-1}\left[\frac{z^2+z}{(z-1)^3}\right] = n^2$, $Z^{-1}\left[\frac{z^3+4z^2+z}{(z-1)^4}\right] = n^3$

4) $Z^{-1}\left[\frac{az}{(z-a)^2}\right] = a^n n$, $Z^{-1}\left[\frac{az^2+a^2z}{(z-a)^3}\right] = a^n n^2$,

$$Z^{-1}\left[\frac{az^3+4a^2z^2+a^3z}{(z-a)^4}\right] = a^n n^3$$

5) $Z^{-1}\left[\frac{z}{z^2+1}\right] = \sin\left(\frac{n\pi}{2}\right)$, $Z^{-1}\left[\frac{z^2}{z^2+1}\right] = \cos\left(\frac{n\pi}{2}\right)$

3.3: Inverse Z-transform by Partial Fraction Method

Procedure: Let $U(z)$ be the given function

Step 1: Write the expression in the form $\frac{U(z)}{z}$.

Step 2: Decompose the expression for $\frac{U(z)}{z}$ into partial fractions.

Step 3: Multiply the partial fraction decomposition by z to obtain $U(z)$.

Step 4: Find u_n by finding $z^{-1}[U(z)]$.

PROBLEMS

1) Find the inverse Z-transform of $U(z) = \frac{2z^2+3z}{(z+2)(z-4)}$.

Answer: $u_n = \frac{1}{6}(-2)^n + \frac{11}{6}4^n$

2) Find the inverse Z-transform of $U(z) = \frac{5z}{(2-z)(3z-1)}$.

Answer: $u_n = -2^n + \left(\frac{1}{3}\right)^n$

3) Find the inverse Z-transform of $U(z) = \frac{18z^2}{(2z-1)(4z+1)}$.

Answer: $u_n = \frac{3}{2}\left(\frac{1}{2}\right)^n + \frac{3}{4}\left(-\frac{1}{4}\right)^n = 3\left[\frac{1}{2^{n+1}} + \frac{(-1)^n}{4^{n+1}}\right]$

4) Find the inverse Z-transform of $U(z) = \frac{4z^2-2z}{(z-1)(z-2)^2}$.

Answer: $u_n = 2 - 2^{n+1} + 3(2)^n n$

3.4: Applications of Z-transform to Difference equations

Introduction: Difference equations can be viewed as the discrete analogue of differential equations. Difference equations, on the other hand, involve rates of change measured by “Discrete generations”. Difference equations are used in variety of contexts, such as in economics, to model the evolution through the time of variables such as gross domestic product, the inflation rate, the exchange rate etc.

Difference Equation: A difference equation is a relation between the differences of an unknown function (dependent variable) at one or more general values of the independent variable. An equation of the form

$$a_0 y_{n+r} + a_1 y_{n+(r-1)} + a_2 y_{n+(r-2)} + \cdots + a_{n-1} y_{n+1} + a_n y_n = \phi(n)$$

where $a_0, a_1, \dots, a_n$ are constants, is called a linear difference equation of order r .

Examples:

- i) $u_{n+1} - u_n = 5$ (linear non homogeneous difference equation of order 1)
- ii) $u_{n+2} + 4u_{n+1} + 3u_n = 3^n$ (linear non homogeneous difference equation of order 2)
- iii) $y_{n+2} - 5y_{n+1} + 2y_n = n^2$ (linear non homogeneous difference equation of order 2)
- iv) $y_{n+3} - 3y_{n+1} + 2y_n = 0$ (linear homogeneous difference equation of order 3)

Working Procedure: To solve linear difference equation with constant coefficients by Z-transform method

- ❖ Take the Z-transform of both sides of the difference equation then use the given initial values and the following formulae
 - i) $Z[u_{n+1}] = z[Z(u_n) - u_0]$
 - ii) $Z[u_{n+2}] = z^2[Z(u_n) - u_0 - u_1z^{-1}]$
 - iii) $Z[u_{n+3}] = z^3[Z(u_n) - u_0 - u_1z^{-1} - u_2z^{-2}]$
- ❖ Transpose all terms without $U(z)$ to the RHS.
- ❖ Divide by the coefficient of $U(z)$, getting $U(z)$ as a function of z .
- ❖ Resolve this function of z into partial fractions and take the inverse Z-transform of both sides. This gives u_n as a function of n which is desired solution satisfying the given conditions.

PROBLEMS

- 1) Use Z-transform method to solve $u_{n+1} - 3u_n = 4$, $u_0 = 1$.

Answer: $u_n = -2(1) + 3(3)^n = 3^{n+1} - 2$.

- 2) Solve $y_{n+2} + 2y_{n+1} + y_n = 0$ with $y_0 = 0, y_1 = 1$, using Z-transform method.

Answer: $y_n = -(-1)^n n = (-1)(-1)^n n = (-1)^{n+1} n$

- 3) Using the Z-transform, solve $u_{n+2} + 4u_{n+1} + 3u_n = 3^n$, $u_0 = 0$ and $u_1 = 1$.

Answer: $u_n = \frac{1}{24} 3^n - \frac{5}{12} (-3)^n + \frac{3}{8} (-1)^n$

- 4) Find the response of the system $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$ with $y_0 = y_1 = 0$, using Z-transform method.

Answer: $y_n = \frac{1}{25} \left[2^n - (-3)^n + \frac{5}{3} (-3)^n n \right]$

Engineering Applications of Z-transform

The Z-transform is a powerful tool that is used in several fields of engineering, including:

Control Systems: The Z-transform is used to model and analyse linear time-invariant systems. By applying the Z-transform to the system's transfer function, engineers can determine the system's stability, response, and performance.

Signal Processing: The Z-transform is used to represent and analyse discrete-time signals and systems. Engineers use the Z-transform to design digital filters and to perform system identification.

Communications: The Z-transform is used to analyse and design communication systems. Engineers use the Z-transform to model and analyse the behaviour of communication channels, and to design error-correcting codes.

Electronics: The Z-transform is used to analyse and design electronic circuits. Engineers use the Z-transform to model and analyse the behaviour of circuits, and to design feedback systems.

Computer Science: The Z-transform is used to analyse algorithms and data structures. Engineers use the Z-transform to model and analyse the behaviour of algorithms, and to design algorithms for specific applications.

Overall, the Z-transform is a versatile tool that is widely used in engineering to model and analyse linear time-invariant systems.

ADDITIONAL PROBLEMS

I. Find the Z-transform of the following functions:

1. 2^{2n+3}
2. $(n+1)(n+2)$
3. $n \cos n\theta$
4. $a^n \cosh n\theta$
5. $\sin(n+1)\theta$
6. $n^2 e^{an}$

II. Find the inverse Z-transform of the following functions:

1. $\frac{z^2-3z}{(z-5)(z+2)}$
2. $\frac{10z}{(z-1)(z-2)}$
3. $\frac{z^2}{(z-3)(z-4)}$
4. $\frac{3z^2+2z}{(5z-1)(5z+2)}$
5. $\frac{z^3}{(z-2)(z-1)^2}$

III. Using Z-transform method solve the following difference equations:

1. $y_{n+1} - 3y_n = 3^n$ with $y_0 = 2$
2. $y_{n+2} - 4y_n = 0$ with $y_0 = 0, y_1 = 2$
3. $u_{n+2} + 4u_{n+1} + 3u_n = 3^n$ with $u_0 = 0$ and $u_1 = 1$
4. $y_{n+2} - 5y_{n+1} + 6y_n = 36$ with $y_0 = y_1 = 0$
5. $y_{n+3} - 3y_{n+1} + 2y_n = 0$ with $y_0 = 4, y_1 = 0$ and $y_2 = 8$
6. $y_{n+2} - 5y_{n+1} + 6y_n = u_n$ with $y_0 = 0, y_1 = 1$ and $u_n = 1$ for $n = 0, 1, 2, 3, \dots$